g_t^L (per layer L)

$m^L \leftarrow \beta_1 m^L + (1-\beta_1) g_t^L$

$v^L \leftarrow \beta_2 v^L + (1-\beta_2)(g_t^L)^2$

$u_adam^L = m^L/(\sqrt{v^L} + \varepsilon) + \lambda\theta^L$

$\varphi^L = \|\theta^L\|_2 / \|u_adam^L\|_2$

$\theta_t^L \leftarrow \theta_{t-1}^L - \eta\varphi^L u_adam^L$